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Oefeningenreeks 6A nr. 7.5

$$\begin{aligned} & \begin{cases} x_1 \cdot x_3 = 36 \\ x_2 + x_4 = 60 \end{cases} \\ & \begin{cases} x_1 \cdot (x_1 \cdot q^2) = 36 \\ x_1 \cdot q + x_1 \cdot q^3 = 60 \end{cases} \\ & \begin{cases} x_1^2 \cdot q^2 = 36 \\ x_1 \cdot q (1 + q^2) = 60 \end{cases} \\ & \begin{cases} x_1 = \pm \frac{6}{q} \\ \left(\pm \frac{6}{q} \cdot q \right) (1 + q^2) = 60 \end{cases} \\ & \begin{cases} x_1 = \pm \frac{6}{q} \\ \pm 6 (1 + q^2) = 60 \end{cases} \\ & \begin{cases} q = \pm 3 \\ x_1 = \pm 2 \end{cases} \end{aligned}$$

$$\Rightarrow x_2 = x_1 \cdot q = 6$$

$$\Rightarrow x_3 = x_2 \cdot q = \pm 18$$

Dus de eerste 3 termen zijn of:

$$x_1 = 2, x_2 = 6, x_3 = 18$$

of:

$$x_1 = -2, x_2 = 6, x_3 = -18$$

References